AI-POWERED CUSTOMER ENGAGEMENT PLATFORM

WhatsApp Automation • AI Voice Agents • CRM & Lead Management

Detailed Technology Stack, Architecture, Delivery Roadmap & Indicative Budget

Prepared for: Smart Software Service

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1. Executive Summary

This document proposes a practical, scalable solution for a business that wants to combine WhatsApp Business messaging, automated customer engagement, lead management, CRM integration, and AI-powered inbound/outbound voice agents. The aim is to respond to enquiries quickly, qualify leads consistently, reduce repetitive manual work, and give sales and support teams a single view of customer interactions.

Recommended approach: deliver a WhatsApp and CRM minimum viable product (MVP) first, then add AI voice calling after the message workflows, data model, consent handling, and reporting are stable. Reuse established APIs and voice-agent platforms rather than building speech recognition, text-to-speech, telephony, or a foundation model from scratch.

Decision area	Recommended starting point	Why
Backend	Node.js + TypeScript + NestJS	Good fit for webhooks, integrations, asynchronous events and a consistent JavaScript/TypeScript stack.
WhatsApp	Meta WhatsApp Business Platform / Cloud API	Direct platform integration and control over message flows.
Workflow automation	n8n	Rapidly connects triggers, CRM actions, notifications and scheduled follow-ups.
Database	PostgreSQL	Relational records for leads, consent, conversations, call outcomes and audit trails.
AI	Provider API such as OpenAI, Gemini or AWS Bedrock	Avoids training and hosting a model at the start.
Voice	Indian-capable telephony provider plus a voice-agent platform	Accelerates inbound/outbound calling; provider capabilities and India support must be validated.
CRM	Integrate the client's existing CRM first	Avoids building a large CRM before requirements are known.
Cloud	AWS managed services	Supports secure deployment, logging, scaling and operational monitoring.

2. Business Requirements and Scope

2.1 WhatsApp API and automation

- Connect a business phone number through the WhatsApp Business Platform.
- Receive customer messages through verified webhooks and send appropriate replies.
- Use approved message templates where required for business-initiated messaging.
- Automate acknowledgements, FAQs, lead capture, routing, reminders and follow-ups.
- Store customer details, message metadata, consent/preferences and interaction history.
- Integrate with the client's CRM or a custom lead-management module.
- Provide failure handling, retry logic, delivery status tracking and operational logs.

2.2 AI voice agents

- Support inbound calls and permitted outbound calls.
- Use speech recognition, a language model and speech synthesis to conduct conversations.
- Collect lead qualification details such as need, location, budget, urgency and preferred contact time.
- Create a call summary, structured outcome and recommended next action.
- Transfer the call or create a human follow-up task when the AI cannot help.
- Track call duration, outcome, transfer rate, latency and cost per completed interaction.
- Apply appropriate consent, recording disclosure, retention and telecom compliance controls.

2.3 CRM and lead management

- Create, assign, search and update leads.
- Maintain a sales pipeline with configurable statuses.
- Link WhatsApp conversations and voice calls to a lead/contact record.
- Record lead scores, summaries, source campaigns and next actions.

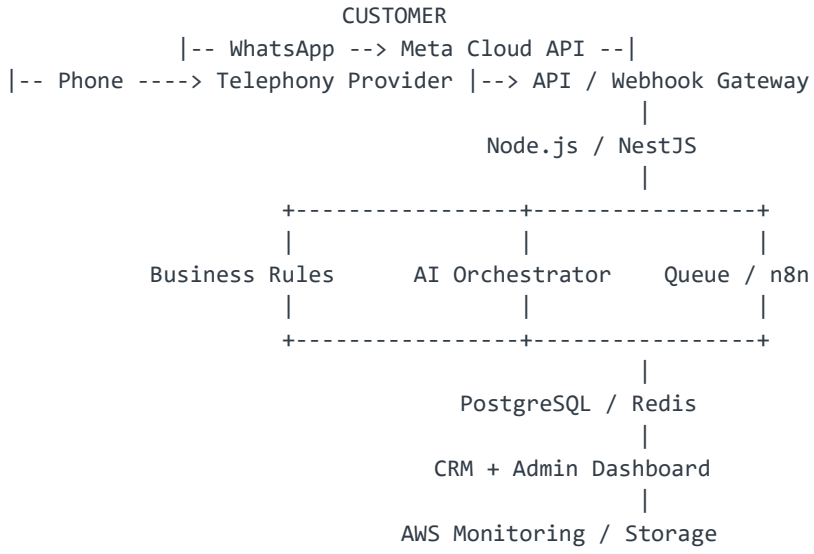
- Support role-based access for administrators, sales representatives and supervisors.
- Provide dashboards for response time, lead qualification, follow-up completion and conversions.

3. Proposed Technology Stack

Layer	Preferred technology	Alternatives / notes
WhatsApp channel	Meta WhatsApp Business Platform / Cloud API	Use the official API; confirm number onboarding, templates, pricing and messaging policy.
API backend	Node.js + TypeScript + NestJS	Express.js is suitable for a smaller MVP; NestJS provides a stronger application structure.
Webhook processing	NestJS controllers + signature/token validation	Validate requests, handle duplicates and acknowledge webhooks promptly.
Workflow automation	n8n	Use for integration workflows; keep critical business rules and authorization in the backend.
Database	PostgreSQL	Store structured leads, consent records, interaction metadata and audit records.
Cache / queue	Redis + BullMQ	Use for retries, scheduled jobs, throttling and asynchronous processing where needed.
AI language model	OpenAI API, Google Gemini or AWS Bedrock	Choose after testing quality, latency, data-handling terms and cost.
Voice agent	Vapi or Retell AI, subject to provider/India suitability	Validate Indian number support, call routing, latency, languages, data residency and pricing.
Telephony	Exotel or Twilio, subject to use case and current availability	Confirm inbound/outbound capabilities, Indian number rules, TRAI-related requirements and commercial terms.
Speech-to-text	Voice-platform native service or Deepgram / Google Cloud	Benchmark English, Hindi, Hinglish and required regional languages.
Text-to-speech	Voice-platform native service, ElevenLabs or Google Cloud	Test voice quality, pronunciation, language support and commercial rights.
CRM	Client's existing CRM; e.g., Zoho CRM or HubSpot	Use supported APIs and webhooks. Build a custom CRM only if justified.
Frontend	React + TypeScript	Optional for MVP; a simple admin view may be enough initially.
Cloud hosting	AWS ECS/Fargate or Lambda based on workload	Use managed services where they reduce operational overhead.
Object storage	Amazon S3	Optional for recordings/transcripts; apply access controls, encryption and retention policies.
Secrets and identity	AWS Secrets Manager / Parameter Store; managed identity provider	Never store API keys in source control; enforce least privilege.
Observability	Amazon CloudWatch + structured logs	Add alarms, correlation IDs, dashboards and error tracking.

4. High-Level Architecture

The proposed architecture separates channel adapters, business rules, AI orchestration, persistence and external integrations.



4.1 WhatsApp message flow

1. A customer sends a message to the registered business number.
2. Meta sends a webhook event to the public HTTPS endpoint.
3. The backend validates the request and acknowledges it quickly.
4. The event is queued or processed asynchronously to avoid webhook timeouts.
5. The application resolves the customer/lead, applies business rules and optionally calls an LLM.
6. The backend sends a response through the Cloud API and records message metadata/status.
7. A workflow can create a CRM task or schedule an eligible follow-up.

4.2 AI voice flow

8. A permitted inbound or outbound call is connected through the chosen telephony/voice-agent platform.
9. Speech is transcribed or processed by the provider's real-time speech pipeline.
10. The AI agent uses a constrained prompt, approved knowledge sources and defined tools.
11. The agent asks qualification questions and extracts structured answers.
12. The system updates the CRM and stores a concise summary and outcome.
13. Calls needing human judgment are transferred or assigned to a person.
14. Monitoring captures failures, latency, call outcomes and usage costs.

5. Suggested Data Model

Entity	Key fields	Purpose
Lead	id, name, phone, email, source, status, owner_id, score, created_at	Core lead record.
ContactPreference	lead_id, channel, consent_status, source, timestamp, opt_out_at	Track channel preferences and consent evidence.
Conversation	id, lead_id, channel, started_at, last_activity_at, status	Group related interactions.
Message	id, conversation_id, direction, provider_message_id, type, status, timestamp	Track inbound/outbound messages and delivery states.
Call	id, lead_id, provider_call_id, direction, duration, outcome, recording_ref	Store call metadata and outcome.
TranscriptSummary	call_id, summary, extracted_fields, model_version	Store useful results without relying only on raw transcript.
FollowUpTask	id, lead_id, due_at, channel, status, assigned_to	Manage scheduled follow-ups.
AuditEvent	id, actor, action, entity, timestamp, correlation_id	Support traceability and operational investigations.

Design note: avoid storing unnecessary sensitive information. Define retention periods and deletion procedures. Store provider identifiers and metadata rather than duplicating large payloads where possible.

6. Security, Privacy and Compliance

- Use HTTPS for all public webhooks and validate provider signatures/tokens as supported by each provider.
- Store API credentials in a managed secrets service and rotate them periodically.
- Apply least-privilege IAM roles and restrict access to recordings, transcripts and customer records.
- Encrypt data in transit and at rest; use database backups and test restoration.
- Record the source, scope and timestamp of customer consent/preferences where required.
- Respect WhatsApp opt-outs, approved template requirements and platform policies.
- For calls, assess applicable Indian telecom, TRAI, Do Not Disturb, consent and recording-disclosure obligations with the provider/legal adviser before launch.
- Define privacy notices, data retention, deletion, access controls and incident response procedures.
- Do not let the AI invent prices, policy promises or eligibility decisions; constrain responses to approved information.
- Provide human handoff for complaints, sensitive requests, low-confidence responses and high-impact decisions.

This document is a technical planning guide, not legal advice. Compliance requirements depend on the call type, industry, customer location, consent model and provider setup. Confirm current requirements before production.

7. Delivery Plan and Milestones

Phase	Scope	Indicative duration	Acceptance criteria
0. Discovery	Confirm use cases, volumes, CRM, languages, consent, call scenarios and success metrics.	3–5 working days	Approved requirements and provider shortlist.
1. WhatsApp MVP	Number setup, webhook, inbound/outbound messages, templates and basic logging.	1–2 weeks	Test messages flow end-to-end and errors are observable.
2. Lead/CRM workflows	Lead capture, CRM sync, status updates, scheduled follow-ups and opt-out handling.	1–2 weeks	Lead records and follow-up tasks are correctly synchronized.
3. AI voice pilot	Inbound/outbound pilot, qualification script, structured extraction, summary and human handoff.	2–4 weeks	Pilot meets agreed language, quality, latency and cost targets.
4. Dashboard and hardening	Reports, access control, alarms, retries, deployment, runbooks and user acceptance testing.	1–2 weeks	Security and UAT checklist signed off.
5. Production rollout	Limited rollout, training, monitoring and post-launch support.	1–2 weeks	Operational owner, support process and rollback plan established.

The timeline is indicative and assumes timely access to Meta/telephony accounts, CRM APIs, test data and client stakeholders. Provider approval or number registration can add lead time.

8. Quality and Acceptance Checklist

- Webhook verification and duplicate-event handling tested.
- Message delivery states and provider errors recorded.
- Approved templates and customer opt-out flows verified.
- Follow-up schedules respect policy constraints and do not contact opted-out customers.
- CRM records remain consistent when a provider request fails or is retried.
- AI voice agent identifies itself appropriately and follows an approved script.
- Low-confidence, complaint and out-of-scope scenarios trigger safe handoff.
- Hindi/English/Hinglish and required regional languages tested with real users.
- Latency, concurrent calls, timeout and provider outage behavior tested.
- Access control, secrets, encryption, backups and retention policies reviewed.
- Dashboard metrics reconciled against provider records.
- Runbooks, rollback procedure and support ownership documented.

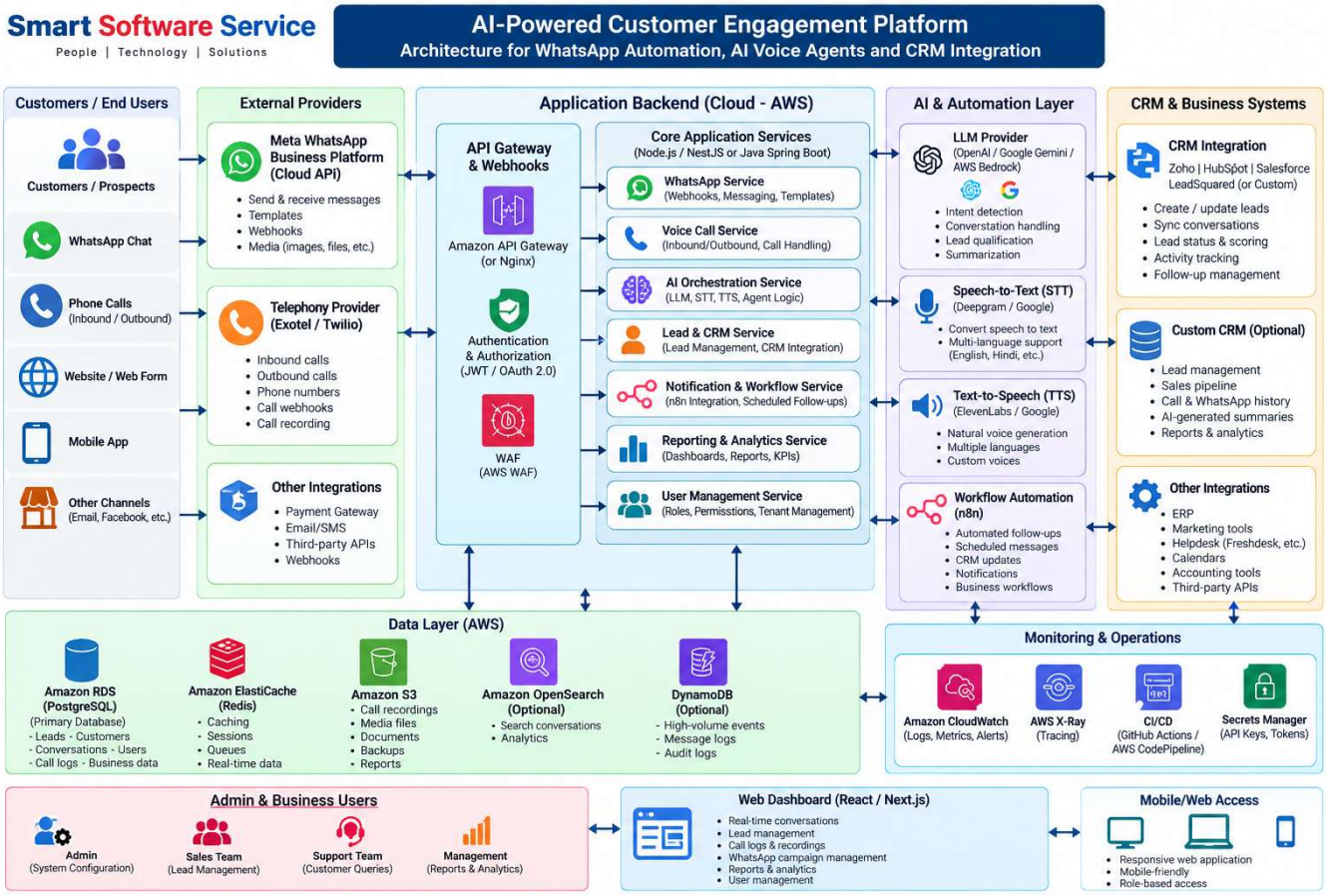
9. Suggested KPIs

Area	KPI	Purpose
WhatsApp	First response time; delivery rate; failure rate	Measure responsiveness and reliability.
Automation	Workflow success rate; retry rate; follow-up completion	Measure operational stability.
Voice	Call connection rate; average latency; completion rate; transfer rate	Measure voice experience and usefulness.
Sales	Lead qualification rate; appointment rate; conversion rate	Measure business impact.
Cost	Cost per qualified lead; cost per completed call	Understand unit economics.
Quality	Human review score; incorrect-answer rate; opt-out/complaint rate	Protect customer experience and trust.

10. Risks and Mitigations

Risk	Mitigation
WhatsApp/telephony onboarding delays	Start account setup and provider checks during discovery; keep a test environment.
Unexpected usage costs	Set usage budgets and alerts; log cost by tenant and workflow; cap call duration where appropriate.
AI hallucinations or wrong commitments	Use approved knowledge, structured outputs, tool allowlists, guardrails and human escalation.
Poor performance in local languages	Benchmark target languages and accents with representative callers before rollout.
Provider outage or rate limits	Queue work, retry safely, use backoff and provide a human/manual fallback.
Privacy or regulatory non-compliance	Complete compliance review, consent/opt-out controls, access restrictions and retention policies.
Scope creep	Define MVP boundaries, acceptance criteria, change control and explicit exclusions.

11. Architecture Diagram



12. Official Technology Resources

- **WhatsApp Cloud API:** https://developers.facebook.com/docs/whatsapp/cloud-api/
- **Meta WhatsApp Developer Hub:** https://whatsappbusiness.com/developers/developer-hub/
- **Node.js:** https://nodejs.org/
- **NestJS:** https://nestjs.com/
- **n8n:** https://n8n.io/
- **PostgreSQL:** https://www.postgresql.org/
- **Amazon Web Services:** https://aws.amazon.com/
- **Amazon CloudWatch:** https://aws.amazon.com/cloudwatch/
- **OpenAI API documentation:** https://platform.openai.com/docs/overview

- **Google Gemini API:** https://ai.google.dev/gemini-api/docs
- **AWS Bedrock:** https://aws.amazon.com/bedrock/
- **Exotel:** https://exotel.com/
- **Twilio Voice pricing for India:** https://www.twilio.com/en-us/voice/pricing/in
- **Vapi:** https://vapi.ai/
- **Retell AI:** https://www.retellai.com/
- **Zoho CRM:** https://www.zoho.com/crm/
- **HubSpot CRM:** https://www.hubspot.com/products/crm
- **ElevenLabs:** https://elevenlabs.io/

13. Questions to Confirm Before Final Quotation

15. Which industry and customer journeys are in scope?
16. Does the customer already have a CRM? If yes, which product and API access level?
17. Expected monthly WhatsApp conversations and outbound message volume?
18. Expected inbound/outbound calls per day, average call duration and peak concurrency?
19. Which languages and accents must the voice agent support?
20. Should the AI agent only qualify leads, or also book appointments, take payments or update orders?
21. Are call recordings required? What retention period and access policy apply?
22. Does the customer have a Meta Business Portfolio, WhatsApp Business Account and suitable phone number?
23. What is the target launch date, support SLA and budget range?
24. Will this be a single-customer deployment or a multi-tenant SaaS product?

14. Recommended Initial Scope

For the first paid pilot, keep the scope controlled: one WhatsApp number, one CRM integration, a small set of approved message templates, lead capture, two or three follow-up workflows, one voice-agent use case, basic dashboard metrics, and a human handoff. Test with a limited customer cohort before scaling.

The best architecture is the one that satisfies the client's business outcomes with the least operational complexity. Start with proven providers, measure usage and conversion, and introduce custom components only where they create a clear advantage.